

Département de Sciences biologiques, Université de Montréal,
Montréal, Québec, Canada H3C 3J7

and

Department of Anatomy, Beijing Second Medical College
Beijing, China

Quantitative Morphology of the Panda Brain in Comparison with the Brains of the Raccoon and the Bear

By

P. PIRLOT and S. S. JIAO*

With 3 Tables

(Received January 4, 1984)

Summary: The quantitative development of brain and brain components with respect to body-size is discussed comparatively for the Chinese Giant Panda, the Canadian Raccoon and the American Black Bear. The panda ranks highest in encephalization as well as in relative volumes of neocortex, cerebellum and six other parts. The raccoon exhibits the strongest development of olfactory elements. The bear stays below the other species, showing but a single index that is superior to the corresponding index in one of them only. These results must be considered within the context of recent debates on the systematic position of the panda. This animal appears to have followed a partly independent evolutionary pathway, although it seems to stand somewhat closer to the raccoon than to the bear, contrary to other views. New comparative eco-ethological studies of the three species, as a complement to our neuromorphological approach, would certainly clarify the status reached by the panda at the end of its long and lonely evolution.

Introduction

While the brains of a number of carnivores have been well described in their qualitative and external morphology, their quantitative aspects and internal organisation have been neglected. Data on the relative size of the brain with respect to that of the body can be found in DUBOIS (1897, 1914), BONIN (1937), WARNCKE (1908), etc, but these data are scanty and relate to total encephalon only. In view of the recently-acquired knowledge of the evolutionary significance of the relative size of the brain and brain components in several other mammalian orders (primates, chiropters, rodents, marsupials), it seems timely to turn to carnivores and, among these, to such a prominent figure as the Giant or Greater Panda from China, *Ailuropoda melanoleuca* David 1869. This animal is now considered an endangered species and great endeavours are being devoted to its conservation. It has indeed become the world symbol of threatened wild-life, and has been very much in the public eye for that reason. Its rarity, its geographic restriction and the strangeness of its bamboo-bound existence have given it an "aura of mystery" (SARICH, 1973) which has increased its attractiveness. But beyond being an elegant black-and-white star in

zoos and the cuddly pet reproduced thousands of times as baby toys, *who* is really the Giant Panda? This basic question remains unanswered. The panda's taxonomic position and systematic relationships are still being disputed.

For a long time, the panda and the raccoon were considered to belong to the same family, the Procyonidae (SIMPSON, 1945). This is at present denied by several authors (HONACKI et al., 1982). Whatever their systematic relationship may be, those two carnivores have long been separated in their history. For some paleontologists, they in fact represent two distinct rami, the Asian Ailurinae (including also *Ailurus fulgens*, the Lesser Panda) and the American Procyoninae. The former are known from the Upper Miocene in Europe and the Lower Pliocene in Asia, where they are now all confined; the latter, from the Lower Miocene in North America and the Upper Miocene in South America (SIMPSON, 1945). Regarding the Giant Panda, SIMPSON (op. cit. p. 226) writes: "Few animals have been of more disputed affinities than *Ailuropoda*". Even its association with *Ailurus* has been successively claimed and disclaimed. Raven and Gregory, independently but in the same year (1936), demonstrated the closeness of their relation with data on the viscera and the skull, as well as

dentition, respectively (in SIMPSON, 1945, p. 226). But again this is now being challenged. One idea which seems to be widely accepted is that both the Ursidae and the Procyonidae derive from the Canidae (op. cit. p. 222). One opinion reported by HONACKI et al. (1982) is that of THENIUS, 1979 who "placed *Ailuropoda* in the monotypic family Ailuropodidae, and *Ailurus* in the subfamily Ailurinae of family Procyonidae".

A very recent work on the taxonomy and geography of mammals (HONACKI et al., 1982), places *Ailuropoda* within the Ursidae and *Ailurus* within the Procyonidae; this could be considered to agree implicitly with SARICH (1973) who showed that, on the basis of immunological distances, the proper order of affinities for the four types at issue should be *Ursus*—*Ailuropoda*—*Ailurus*—*Procyon*. But, as reported by HONACKI et al. (1982), the exact location each genus is still a matter of opinion since other recent authors continue to accept a separate family the Ailuropodidae, either for *Ailuropoda* alone or for *Ailuropoda* and *Ailurus* jointly.

We wish to contribute to the panda problem by providing some basic quantitative data on the brain and brain-components of this Chinese species. This should help establish the panda's status among mammals and lead to a better understanding of its ecology and behaviour. The circumstantial cause of this undertaking was the opportunity that arose when the first author visited the Beijing Second Medical College in 1980. It was then agreed with the researchers of that institution that such a study, within the framework of a comparison with the Canadian black bear *Ursus americanus* Pallas 1780, and raccoon *Procyon lotor* L., would be of considerable value. A plan for such a collaboration was eventually approved by the Chinese authorities at all levels, the Université de Montréal and the Natural Sciences and Engineering Research Council Canada. The first result of this collaboration is the present paper which will be followed by at least two more on the same material.

Materials, Techniques and Methods

One brain only from each species was studied.

- 1) *Ailuropoda melanoleuca*. Giant panda. Male. Age 8–9 years. Obtained in Beijing, People's Republic of China, by the second author. Body-weight: 37.5 kg. Brain-weight: 226.0 g.
- 2) *Procyon lotor*. Raccoon. Female. Age unknown. Obtained in Montréal, Canada, by the first author, through the courtesy of Prof. J. Ferron, Université du Québec à Rimouski. Body-weight: 4.1 kg. Brain-weight: 37.5 g.
- 3) *Ursus americanus*. Black bear. Male. Juvenile. Obtained in Montréal, Canada, by the first author, through the courtesy of the director of the zoological garden, Parc Angrignon. Body-weight: 49.4 kg. Brain-weight: 149.9 g.

The bear and the raccoon were perfused through the left ventricle with formalin (1/7 water). For the panda, which was not collected by us but whose brain was supplied by the Chinese Authorities in a zoological garden, we have no details about technical procedures. The panda's name was Wei-wei (a boy's name) while it lived in the zoo. The brains were embedded in paraplast and sectioned at 15 μ m, 10 μ m and 15 μ m respectively. All sections were Nissl-stained. Photograms were obtained for each brain by projecting directly onto paper a series of about 125 sections, regularly spaced. The main neuronal fields were delimited on the pictures so produced and their areas measured with an electronic image analyzer (Zeiss Videoplan). By simple computation and correction for contraction of the tissues, the original volume of each brain component was estimated.

The comparisons among the three brains are partly based on the allometric method of progression indices. This method has been explained in several publications (ex. STEPHAN 1967; PIRLOT and KAMIYA 1982). From the latter paper (p. 566), one may quote: "The principle of the allometric comparisons is simple: (1) the classical allometry equation $y = bx^k$ relating brain size (y) to body size (x) is applied in its log form $Y = B + kX$ where $Y = \log y$, $B = \log b$ and $X = \log x$, while k is the growth exponent; (2) a composite group called the "basal insectivores" by STEPHAN (1967) is taken as reference i.e. the constants of the equation $Y = B + kX$ for that group are used to estimate the brain sizes of basal insectivores whose body sizes would be equal to those of the animals under investigation; (3) the ratios of the actual brain sizes in the animal studied to the expected brain sizes in the corresponding basal insectivores are taken as progression indicators, i.e. they express how many times the brains under investigation are larger than the brains of those hypothetical insectivores of same body size; (4) the same procedure is applied to the main brain components as delimited on photographs of brain sections. Multiplying the ratios (actual/expected) by 100, one obtains progression indices which may vary from below 100 to above 100, the 100 level indicating the theoretical expected value in the reference insectivores".

In parallel with the progression index method, a simultaneous approach to comparisons was used, based on the percentage value of each brain part with respect to total brain and of each telencephalic component to total telencephalon in each species.

Results

We shall first consider the total brain and then the brain composition.

Total brains

The progression indices obtained from the brain-weight to body-weight ratios are 692 for *Ailuropoda*, 463 for *Procyon* and 386 for *Ursus*. To make these values more meaningful to the reader, the following previously published figures can be quoted: 2881 for *Homo*, 1182 for *Pan* (chimpanzee), 417 for *Galago* (bush-baby, prosimian), calculated from data in Stephan, Bauchot and Andy, 1970; 609 to 2308 in dolphins (Piriot and Kamiya 1975); 95 to 275 in bats (PIRLOT and POTTIER 1977); 92 to 383 in rodents

(PIRLOT and KAMIYA 1982, and PIRLOT and SPERONI, in preparation). Within this set of data, the values obtained for our three carnivores are at a "medium-level".

Brain components

The ratios of the 10 brain components studied to total brain for each species are given in Table 1; those of some telencephalic parts to total telencephalon in Table 2. The progression indices for all components are presented in Table 3.

Table 1. Percentile composition of total brain.

	Panda	Raccoon	Bear
BO	0.61 (1306.25)	2.16 (774.60)	1.12 (1569.88)
N	64.42 (136435.71)	58.54 (20987.17)	63.00 (88420.25)
RH	3.08 (6515.95)	6.39 (2293.44)	3.73 (5222.64)
S	0.30 (636.39)	0.27 (98.97)	0.44 (625.64)
D	4.99 (10559.60)	6.21 (2224.59)	4.82 (6764.36)
St	5.99 (12698.38)	6.18 (2213.12)	6.69 (9383.66)
H	2.21 (4694.22)	3.45 (1240.37)	1.74 (2446.69)
M	1.87 (3970.35)	2.80 (1006.39)	2.41 (3389.46)
C	14.64 (30994.23)	11.18 (4005.21)	13.78 (19339.75)
O	1.89 (3998.28)	2.82 (1012.90)	2.27 (3185.25)
Total	100.00 (211809.35)	100.00 (35856.76)	100.00 (140347.58)

BO = bulbus olfactorius; N = neocortex; RH = rhinencephalon; S = septum; D = diencephalon; St = striatum; H = hippocampus; M = mesencephalon; C = cerebellum; O = medulla oblongata. In brackets, the absolute volumes in mm³.

Table 2. Percentile composition of telencephalon

	Panda	Raccoon	Bear
BO	0.80	2.80	1.45
N	84.07	76.02	82.13
RH	4.01	8.31	4.85
S	0.39	0.35	0.58
St	7.82	8.02	8.72
H	2.89	4.50	2.27

Abbreviations as in Table 1.

Table 3. Progression indices of brain parts

	Panda	Raccoon	Bear
BO	38	93	38
N	3018	1957	1635
RH	140	187	96
S	162	91	136
D	478	397	258
St	1027	690	641
H	259	242	116
M	311	278	227
C	613	357	317
O	136	122	93

Abbreviations as in Table 1.

Discussion

A. Encephalization

Both the high figure for the panda (692) and the low figure for the bear (386) are somewhat surprising, especially since the bear specimen used for this work was a very young animal and it is known that relative brain weight is usually higher in young than in old individuals, owing to negative ontogenetic allometry. From data on five more pandas collected by the second author in Beijing, we can derive some comparative values. Pooling these five sets of figures with ours, we obtain for the six animals (4 males and 2 females) a mean brain weight of 260.8 g and a mean body weight of 61.5 kg. These figures yield an average progression index of 585. The heaviest of these six pandas, a male weighing 89 kg, gave an index of 576. It is clear that the specimen we used in the present work had a relatively very high index; this may be due to its being relatively young and light; thus, the high value obtained should be interpreted with caution.

It would certainly be interesting to present a substantial series of progression indices for other ursids and procyonids. We unfortunately do not possess such data at this time. Two figures only can be given. One is for an individual of *Nasua* (the South American coati, a procyonid) in Pirlot's collection, which yields a progression index of 631. This is a rather high figure, if we compare it with the indices for the North American raccoon (463) and the average panda (585). The other one is for the species *Helarctos malayanus*, a South-east Asian bear mentioned in Jerison (1973, p. 359), for which the surprisingly high index of 1065 was obtained.

The meaning of encephalization as an indicator of progressive evolution has been the object of much speculation from HALLER (1762) to DUBOIS (1897) in past generations, and to JERISON (1973) in our lifetime. In spite of these lengthy debates, we are still

fairly uncertain as to how total brain size is related to intelligence, skill and survival in wild animals. The mere variety of BrW/BoW ratios in domestic dogs is puzzling and the figures for the Canidae in general are not very revealing. It must be recalled here that the Ursidae and the Procyonidae are considered to be derived from Cenozoic canids. People who live in bear and raccoon areas in Canada and the United States would probably rank the raccoon among the "smart" animals around but credit the bear with a lesser degree of cleverness, which would be a subjective judgment, to be sure. Breeding and training bears can be quite successful as anyone knows who has visited a travelling circus. But what can one really conclude from such training as produces a pet raccoon or a clown bear, as far as the relative degree of intelligence in the two animals is concerned? JERISON's (1973) book alone is proof that this avenue of thinking is fraught with treacherous snags, and the current dissents over the educational level really attained by "talking" chimpanzees do not point to the existence of an objective set of criteria for evolutionary progress related to brain size in sub-human mammals. In any case, taking our figures 386 and 463 as acceptable indicators of such progress in the bear and the raccoon respectively, we can only hypothesize as to why the panda stands so much higher than the other two carnivores (1.79 and 1.49 times higher if we take the index 692 for our panda alone, 1.51 and 1.26 times if we take the average index 585 from the group of six). This "superiority" is not likely to be explained until we know more about the ecology and ethology of the species in the wild. This, in fact, applies even more cogently to the Malayan bear, for which we had second-hand data only and which certainly deserves detailed brain analysis.

B. Brain composition

1) Progression indices. As would be expected from the encephalization figures, the panda shows the highest neocortical index among our three animals, i.e. 3018 against 1957 for the raccoon and 1635 for the bear. The panda/bear ratio for their two indices is almost 2/1, which is remarkable in animals that are supposedly fairly close to one another. It could be said that the panda's evolutionary superiority in this respect (i.e. for the brain-to-body size relationship) is well demonstrated. In comparison, PIRLOT and KAMIYA (1980) found 3903 for *Platanista*, a relatively lowly dolphin. Some prosimians (e.g. *Tarsius*, *Galago*) stand much lower than the panda on the neocortical scale. Again, most people would probably find it difficult to understand why the panda should be so high.

The panda's cerebellar index also is the highest of all — and it is known that the cerebellum often follows the neocortex in its progression trends (STEPHAN and PIRLOT 1970; PIRLOT and POTTIER 1977). The figure is 613 for the panda against 357 for the raccoon and 317 for the bear. Cerebellar progression is often thought of as related to superiority in climbing ability. If that is so, the panda would be expected to have a high cerebellar index; but the raccoon and the bear too are good climbers and thus, the difference between them is, again, difficult to explain. Considering the manual skill of these animals, however, the very high index found for the panda would be consistent with the possibility that the use of a "sixth finger" (the panda's thumb) requires increased cerebellar control; this, however, remains to be proved. According to a number of reports, the raccoon too possesses a special skill in its thumb, and the sensory representation for that finger in its brain has been found to be exceptionally large (HERRON 1978). But then the large difference between panda and raccoon on the one hand, and the absence of large differences between raccoon and bear on the other hand, are surprising. Perhaps the bear is about as clever as the raccoon with its forefingers.

The olfactory structures are definitely more progressive in the raccoon than in the other two species. However, the progression of the panda's rhinencephalon is close to the raccoon's. The main difference is in the bulb, which is almost non-regressive (93) in the raccoon, whereas it is strongly and equally (38) regressive in the panda and the bear.

The striatum is markedly more progressive (one and a half times) in the panda than in the raccoon and the bear. A great variation in the progression of the striatum has been noted by PIRLOT and POTTIER (1977) and STEPHAN and PIRLOT (1970); the latter authors concluded that "There are obviously narrow size-relationships between striatum and neocortex which were also observed in primates" (p. 213). Our findings here agree quite well with that statement and also with a more accurate analysis carried out on a variety of bat brains by PIRLOT and JOLICOEUR (1982).

The hippocampus is about equally progressive in the panda and the raccoon but it is less than half as progressive in the bear. It will be recalled that, in the bear, the olfactory components also are much less progressive than in the other two species.

The indices for the diencephalon present the same relative positions as the neocortex in these three mammals. A corresponding strong correlation between the progressions of the two components was found in bats by PIRLOT and JOLICOEUR (1982).

2) Percentile development. In contrast to the phylogenetic progression with respect to body-size, the relative size of each component within the brain itself gives information as to the relative importance of each function. Telencephalon and total brain are taken in turn as reference values. The relative volume of the neocortex is slightly higher in the panda than in the bear and in the latter than in the raccoon: these differences may not be significant in terms of "cleverness", in contrast to the differences in progression indices which indicate significant increases in brain size with respect to body size throughout historical evolution. The striatum does not differ much in within-brain relative size from one species to the other. The most telling figures are those for the olfactory bulbs and the rhinencephalon, which suggests a most intensive function in the raccoon. Similarly, and rather logically, the raccoon also has the largest share of hippocampus. The cerebellar percentage is the highest in the panda, but the variation among the three animals is not great. So, considered in themselves, and not in relation to the body, the three brains exhibit a fairly similar quantitative composition: this obviously agrees with the similarities observed in the behaviour of the three carnivores under study.

Conclusions

The quantitative analysis of the brains, reported here, throws some light on the panda's position with respect to the raccoon and black bear.

a) As far as the allometric relationship between brain size and body size is concerned, the main observation is that of an evolutionary trend towards (1) higher encephalization and neocorticalization, (2) higher cerebellar progression in the panda than in the other two species. This by itself singles out the panda and agrees with the description of external brain features in the same three animals provided by XIE, PIRLOT and JIAO (in press). Further, from that viewpoint, the panda stands closer to the raccoon than to the bear, in contrast to what we and other authors have suggested on the basis of superficial morphology. These remarks are relevant to the phylogeny of the three carnivores here studied.

b) Concerning the general internal organisation of the three brains, there is a fair degree of uniformity. The neocortex, however, is largest in the panda, which tends to confirm, from a functional standpoint, the idea of a certain superiority of the brain of that animal. The cerebellum is also largest in the panda but, from what is known about the life habits of the three mammals, the eco-ethological significance of this observation is unclear.

c) Regarding the special senses, the most significant finding seems to be the greater development of the olfactory brain components (bulbs and rhinencephalon) in the raccoon.

d) The general conclusion is that the Giant Panda (*Ailuropoda*) must be considered as having followed a distinct evolution in comparison with the raccoon (*Procyon*) and the bear (*Ursus*), with respect to the two or three most important brain components as well as total brain. This conclusion agrees with those of our study of the external anatomy of the same brains (XIE, PIRLOT and JIAO, in press). The addition of our quantitative neuromorphological data to those obtained from osteological, splanchnological and immunological investigations, as well as from ecological and ethological studies, should enable taxonomists to suggest, with better arguments than in the past, what the precise status (familial or subfamilial) of the Giant Panda really is.

NOTE

While this paper was being readied for publication, the authors received a couple of short articles in which other neurobiologists introduce a metabolic coefficient in the brain/body size equation. This is a very interesting approach, *not at variance* with the one taken here, and one that deserves a full-length discussion in methodology as demonstrated by the debate initiated on that problem at the recent International Symposium in Vertebrate Morphology in Giessen, West Germany (August 1983), where one of us (P.P.) was a participant. Obviously, we cannot enter here into such a discussion which could double the size of the present paper if it is to be significant at all. The first author is dealing with it in a review now in preparation for the *Journal für Hirnforschung*.

Acknowledgements

We wish to thank the Chinese scientific and civilian authorities for accepting the proposal from the first author to start this cooperative project, especially Prof. G. B. LEE, Vice-President, Beijing Second Medical College. Mrs. H. RIETVELD-FERNANDEZ gave valuable advice to help solve histological problems. Mrs. J. POTTIER was good enough to check the English text. Support for this work was received from the Université de Montréal (Rector's Funds) and the Natural Sciences and Engineering Research Council Canada (NSERC), which jointly financed the passage to Canada and the stay in Montreal of S. S. JIAO. Other expenses were also defrayed by the NSERC in the form of research grant A-0778 to P. PIRLOT. We dedicate this work to the memory of Prof. J. Q. XIE, former director, Department of Anatomy, Beijing, who helped us throughout but died while the project was being completed.

Literature

- BONIN, G. von: Brain weight and body weight in mammals. *J. Gen. Psych.* 16, 379–389 (1937).
DUBOTS, E.: Ueber die Abhängigkeit des Hirngewichts von

- der Körpergrösse bei den Säugetieren. Arch. Anthropol. 25, 1–28 (1897).
- DUBOIS, E.: Die gesetzmäßige Beziehung von Hirnmasse zur Körpergröße bei den Wirbeltieren. Z. Morph. Anthropol. 18, 323–350 (1914).
- HALLER, A. von: Elementa Physiologiae corporis humani. Lausanne (1762).
- HERRON, P.: Somato-topic organization of mechano-sensory projections to SII cerebral neocortex in the raccoon (*Procyon lotor*). J. Comp. Neurol. 181 (4), 717–727 (1978).
- HONACKI, J. H., K. E. KINMAN and J. W. KOEPL: Mammal Species of the World. Allen Press, Lawrence (1982).
- JERISON, H. J.: Evolution of the brain and intelligence. Academic Press, New York (1973).
- PIRLOT, P. and P. JOLICOEUR: Correlations between major brain regions in Chiroptera. Brain, Behav. Evol. 20, 172 to 181 (1982).
- PIRLOT, P. and T. KAMIYA: Comparison on ontogenetic brain-growth in marine and coastal dolphins. Growth 39, 507–524 (1975).
- PIRLOT, P. and T. KAMIYA: Brain organization in *Platanista gangetica*. Sci. Rep. Whales Res. Inst. 32, 105–126 (1980).
- PIRLOT, P. and T. KAMIYA: Relative size of brain and brain components in three gliding placentals (Dermoptera-Rodentia). Can. J. Zool. 60, 565–572 (1982).
- PIRLOT, P. and J. POTTIER: Encephalization and quantitative brain composition in bats in relation to their life habits. Rev. Can. Biol. 36, 321–336 (1977).
- PIRLOT, P. and N. BEE de SPERONI: Quantitative brain organization in South-American rodents (In preparation).
- PIRLOT, P. and H. STEPHAN: Encephalization in Chiroptera. Can. J. Zool. 48, 433–444 (1970).
- SARICH, W. K.: The giant panda is a bear. Nature 245 (5422), 218–220 (1973).
- SIMPSON, G. G.: The principles of classification and a classification of mammals. Bul. Amer. Mus. Nat. Hist. 85, 1–350 (1945).
- STEPHAN, H.: Zur Entwicklungshöhe der Insektivoren nach Merkmalen des Gehirns und die Definition der „Basalen Insektivoren“. Zool. Anz. 179, 177–199 (1967).
- STEPHAN, H., R. BAUCHOT and O. J. ANDY: Data on the size of the brain and of various brain-parts in Insectivores and Primates. In: NOBACK, C. R. and W. MONTAGNA, eds., The Primate Brain. Appleton, Century, Crofts: 289–297 (1970).
- STEPHAN, H. and P. PIRLOT: Volumetric comparisons of brain structures in bats. Z. zool. Syst. Evol.forsch. 8, 200–236 (1970).
- WARNCKE, P.: Mitteilung neuer Gehirn- und Körpergewichtsbestimmungen bei Säugern. J. Physiol. Neurol. 13, 355–403 (1908).
- XIE, J. Q., P. PIRLOT and S. S. JIAO: Comparative external morphology of the brains of the Giant Panda, the Raccoon and the American Bear (in press).

Send offprint requests to:

P. PIRLOT, Prof.
Département de Sciences biologiques
Université de Montréal
Montréal, Québec, Canada H3C 3J7